

ECE422 — MIDTERM PREP

Homework-style problems + EX3 (Jeopardy) review · Ch.1 Intro, History, Protocols · Ch.2 Encoding, Framing, Error Detection, Ethernet, CSMA, Wi-Fi

Name: _____

Date: _____

Score: _____

Conventions used by the professor (from his HW2 solution and slides):

- **Manchester:** 0 = low → high, 1 = high → low (= NRZ XOR clock). He draws a CLK line above NRZ and Manchester.
- **2-D parity corner bit:** parity of the **P column** (the row-parity bits). HW2: P column 0,0,1 → odd parity → corner **0**.
- **Checksum:** add two words → add the third → add the carry back ("plus carry") → complement.
- **Counting errors:** written as $n!/(k!(n-k)!)$ — e.g. $5!/(3!2!) = 10$.
- **CRC answers** may be written with a leading 0 (EX3: "R = 0111" for a 3-bit remainder 111).

PART I — FORMULA & FACT SHEET

Calculations

Topic	Formula / rule	Example
Shannon	$C = B \cdot \log_2(1 + S/N)$; $dB = 10 \cdot \log_{10}(S/N)$	$B = 5000$, $S/N = 15 \rightarrow 5000 \cdot \log_2 16 =$ 20,000 b/s (EX3)
Min frame to detect collision	$L = 2 \cdot d \cdot R / c$ (transmit for one round trip)	$d = 2500$ m, $R = 10$ Mb/s, $c = 2 \times 10^8 \rightarrow$ 250 bits (EX3)
Ethernet 512 bits	round trip $51.2 \mu s \times 10$ Mb/s	= 512 bits = 64 bytes (14 hdr + 46 data + 4 CRC)
Exponential backoff	after n collisions pick $k \in \{0 \dots 2^n - 1\}$, wait $k \times 51.2 \mu s$	$n = 2 \rightarrow 0, 51.2, 102.4, 153.6 \mu s$
Wavelength	$\lambda = \text{speed} / f$; copper speed $\approx 2/3 c$	300 Hz on copper $\rightarrow 667$ km
Overhead	headers / (headers + data)	4 layers $\times$ 20B, 1B data $\rightarrow 80/81$
k-bit errors in n bits	$C(n,k) = n!/(k!(n-k)!)$	$C(5,2) = 10$
CRC	append r zeros ($r = \text{degree} = \text{bits of } C(x) - 1$), XOR-divide, remainder = CRC	$C(x) = 100 = x^2 \rightarrow R = 0$ iff last 2 bits are 0
MAC $\rightarrow$ decimal	read the hex digits as one number	00:00:00:00:0A:00 = $0x0A00 = 2560$ (EX3)

Ethernet (802.3)

- Xerox PARC, mid-1970s; root protocol **ALOHA** (U. of Hawaii, medium = air). DEC + Intel + Xerox 10 Mbps standard 1978 $\rightarrow$ IEEE 802.3; Fast Ethernet 100 Mbps, Gigabit 1000 Mbps.
- **CSMA/CD:** carrier sense (tell idle from busy) + collision detection (listen while transmitting). Uses **Manchester** encoding.
- Coax segment ≤ 500 m (50Ω), ≤ 4 repeaters between hosts, total reach **2500 m**. 10Base5 (original), **10Base2** = 10 Mbps, baseband, 200 m; **10BaseT** = twisted pair, 100 m, hub.

- **Frame:** preamble 64 bits · dst 48 · src 48 · type 16 (demux key) · data 46–1500 bytes · CRC 32.
- **Address:** 48 bits, belongs to the adaptor (ROM), written as 6 hex bytes, leading 0s dropped (8:0:2b:e4:b1:2). Manufacturer prefix 24 bits (AMD = 8:0:20). All 1s = **broadcast**; first bit 1 but not all 1s = **multicast**. Adaptor accepts: own address, broadcast, multicast it was told to accept.
- **Transmitter:** idle → send now; busy → wait until idle then send at once = **1-persistent**. Collision → send **32-bit jam** then stop. Runt frame = 64 preamble + 32 jam = **96 bits**. Retry with **exponential backoff**.
- Works best lightly loaded; typically < 200 hosts (max 1024); easy, cheap, no routing tables.

Random access (CSMA notes)

Protocol	Channel idle	Channel busy
Pure ALOHA	Transmit at arbitrary times — no carrier sense . Vulnerable period = 2 frame times.	
1-persistent CSMA	transmit (probability 1)	keep listening, transmit as soon as idle
Non-persistent CSMA	transmit	wait a random time, start again
p-persistent CSMA (slotted)	transmit with prob. p, defer to next slot with q = 1-p	wait a random time, start again

All: collision → wait a random time and start over. Data link layer = **LLC** + **MAC** sublayers.

Wireless / 802.11 / Bluetooth / ZigBee

- FCC allocates bands. License-exempt bands: limited **power** + must use **spread spectrum: frequency hopping** (pseudorandom sequence of frequencies, same seed at receiver) or **direct sequence** (each bit XOR n random bits = n-bit **chipping code**, signal n× wider).
- 802.11 original: FH over 79 × 1-MHz, or DS with 11-bit chipping; 2.4 GHz, 2 Mbps. **802.11b** 11 Mbps (2.4 GHz). **802.11a** 54 Mbps, OFDM, 5 GHz. **802.11g** 54 Mbps, OFDM, 2.4 GHz, backward compatible with b.
- **Hidden node:** A and C both reach B but not each other → their frames collide at B and neither knows. **Exposed node:** C hears B → A and wrongly thinks it cannot send to D.
- **MACA:** sender sends **RTS** (with length) → receiver answers **CTS**. Node hearing CTS → stay quiet. Node hearing RTS only → free to send. MACAW adds **ACK**. No collision detection: missing CTS → collision → exponential backoff.
- **Distribution system:** Access Points (AP) wired together (layer 2). **Active scanning:** Probe → Probe Response → Association Request → Association Response. **Passive scanning:** AP sends **Beacon**. New AP tells old AP.
- **Frame:** 4 addresses, ToDS/FromDS bits, data ≤ 2312 bytes, CRC 32, control 16 bits (6-bit type). Both DS = 1: Addr1 = ultimate destination, Addr2 = immediate sender (last AP), Addr3 = intermediate destination (first AP), Addr4 = original source.
- Mobility: none (early WiMAX), within one base (Bluetooth), between bases (cell, Wi-Fi). Mesh / ad-hoc = peers forward for each other.
- **Bluetooth:** 2.45 GHz, ~10 m, PAN, v2.0 = 2.1 Mbps, low power; **piconet** = 1 master + up to 7 slaves, slaves talk only via master, can be **parked**. **ZigBee:** IEEE 802.15.4, lower bandwidth, lower power, cheaper.

Earlier topics (still on the midterm)

- TCP/IP (slides): Application, Transport, Network, Link — routers have the bottom two. OSI has Physical, Session, Presentation that TCP/IP does not (EX3).
- Transport layer: reliable message delivery, flow control, error recovery. MAC (hardware) address only travels to the **closest router**; each hop uses the **next hop's** HW address. Same LAN ⇔ same **network ID**

(masked address).

- Packet switching gives resiliency via metadata + routing/forwarding; Sputnik (1957) → ARPA → communication tolerant to partial destruction. ARPANET 1969: UCLA, SRI, UCSB, Utah. Killer apps: e-mail, WWW.
- Encoding: NRZ, NRZI (flip on 1), Manchester (bit rate = $\frac{1}{2}$ baud rate), 4B/5B (≤ 1 leading 0, ≤ 2 trailing 0s, 80%). FSK = different frequency for 0 and 1.
- Error detection: parity misses even # of errors; 2-D parity catches 1/2/3-bit; checksum (IP, not link level); CRC (HDLC, DDCMP, CSMA/CD, Token Ring).

PART II — HOMEWORK-STYLE PROBLEMS

A. Encoding & Shannon

1. Draw CLK, NRZ, NRZI (line starts low) and Manchester for **11001010**.

2. Show an example of FSK for **10011**.

3. Max data rate: (a) $B = 8000$ Hz, $S/N = 31$; (b) $B = 3000$ Hz, $SNR = 20$ dB; (c) radio link 40 kHz, signal 100 mW, noise 10 mW.

4. Which 2B/3B codes guarantee no more than two consecutive **1s** in the bit stream? Give a valid 2B $\rightarrow$ 3B table.

B. Parity

5. Show the 2-D parity of **1011**, **0110**, **1100** using **odd** parity (use the professor's corner rule).

6. Show the 2-D parity of **1100101**, **0011010**, **1110001** using **even** parity.

7. Which two-bit errors are not detected for **0110P**, $P = \text{odd parity bit}$? List all strings.

C. Checksum

8. Show the 8-bit checksum transmitted to protect the fields **3f:c8:91** (hex). Verify at the receiver.

9. Show 3 field values that produce the same checksum as Q8.

D. CRC

10. $M = 11010011$, $C(x) = 1011$. Find R and the transmitted frame.

11. The receiver of Q10 gets **11010001011**. Is there an error?

12. If $C(x) = 1000$ for a 6-bit $M(x)$, how many nonzero $M(x)$ give $R(x) = 0$? List them.

E. Ethernet / MAC

13. $d = 1000$ m, $R = 100$ Mb/s, $c = 2 \times 10^8$ m/s. Minimum frame length L to detect a collision? Also show why classic Ethernet uses 512 bits.

14. An adaptor has collided 3 times. Which waiting times can it choose? What is the maximum?

15. (a) Write 8:0:2b:e4:b1:2 in binary. (b) What decimal number is 00:00:00:00:01:2C? (c) Which frames does an adaptor accept?

16. A packet goes from host H to a destination 3 hops away ($H \rightarrow R1 \rightarrow R2 \rightarrow D$). Give the destination HW address on each hop. How far does H's HW address travel?

F. CSMA & Wi-Fi

17. Name the protocol: (a) channel busy $\rightarrow$ keep listening and send the moment it is idle; (b) channel busy $\rightarrow$ wait a random time; (c) idle $\rightarrow$ send with probability p; (d) no carrier sense at all.

18. Nodes A–B–C–D in a line, each reaches only its neighbours. (a) A and C both send to B — what problem? (b) B sends to A, C wants to send to D — what problem? (c) How does RTS/CTS help?

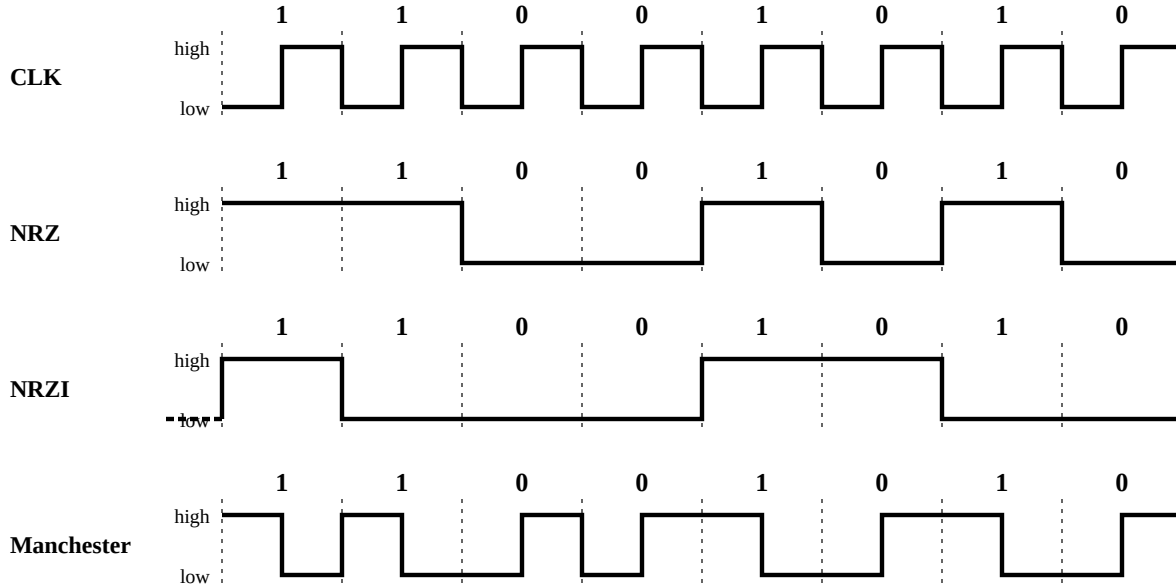
19. Node C (cell of AP-2) sends a frame to node F (cell of AP-1) over the distribution system. Give ToDS, FromDS and Addr1–Addr4.

G. Layers

20. Each layer of the TCP/IP model adds a 20B header. Overhead ratio for 1B and 100B of data?

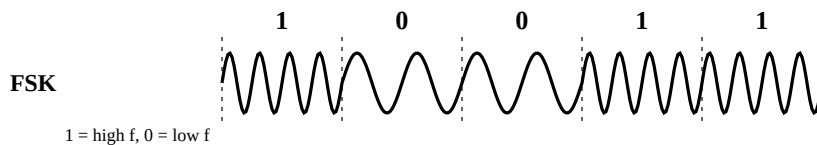
PART III — ANSWER KEY

1. 11001010 — Manchester = NRZ XOR CLK (0 = low → high, 1 = high → low).



NRZI: flip on each 1 → H L L L H H L L.

2. FSK: amplitude constant, a **higher frequency for 1**, a **lower frequency for 0**.



3. (a) $8000 \cdot \log_2(32) = 8000 \times 5 = \mathbf{40,000 \text{ b/s}}$. (b) $20 \text{ dB} \rightarrow S/N = 100 \rightarrow 3000 \cdot \log_2(101) \approx 3000 \times 6.66 \approx \mathbf{20 \text{ kb/s}}$. (c) $S/N = 100/10 = 10 \rightarrow 40,000 \cdot \log_2(11) \approx 40,000 \times 3.46 \approx \mathbf{138 \text{ kb/s}}$.

4. Reject codes that start with 11, end with 11, or contain 111 (011, 110, 111) → valid: **000, 001, 010, 100, 101** (5 codes; need 4). At a boundary at most 1 trailing + 1 leading 1 meet → never more than two 1s in a row. Table: 00 → 000, 01 → 001, 10 → 010, 11 → 100 (101 spare).

EX3 version (no more than two consecutive 0s): 010, 011, 101, 110, 111.

5. Odd parity:

	Data	# 1s	P
R1	1 0 1 1	3	0
R2	0 1 1 0	2	1
R3	1 1 0 0	2	1
Parity	1 1 1 0		1

Corner (professor's rule): P column 0,1,1 has two 1s → odd parity → **1**. (The parity row 1110 would give 0 — the two methods disagree with odd parity, so follow his rule and say so.)

6. Even parity:

	Data	# 1s	P
R1	1 1 0 0 1 0 1	4	0
R2	0 0 1 1 0 1 0	3	1
R3	1 1 1 0 0 0 1	4	0
Parity	0 0 0 1 1 1 0		1

With even parity both corner methods agree (= 1).

7. 0110 has two 1s $\rightarrow P = 1 \rightarrow$ **01101**. All 2-bit errors are missed: $5!/(2!3!) = 10$:

Flipped	1,2	1,3	1,4	1,5	2,3	2,4	2,5	3,4	3,5	4,5
String	10101	11001	11111	11100	00001	00111	00100	01011	01000	01110

8.

$3f = 0011\ 1111$ $+ c8 = 1100\ 1000$ Sum 1 = 1 0000 0111 (carry out) $+ 91 = 1001\ 0001$ Sum 2 = 1 1001 1000 $\rightarrow$ plus carry: $1001\ 1000 + 1 = 1001\ 1001$ (0x99) Complement $\rightarrow 0110\ 0110 = 0x66$ $\leftarrow$ transmitted checksum Receiver: $3f + c8 + 91 + 66 \rightarrow 1111\ 1111 \rightarrow$ complement $0000\ 0000 \rightarrow OK$

9. Any fields with the same ones-complement sum (0x99): **91:c8:3f** (reordered), **40:c7:91** (+1/-1), **99:00:00**.

10. $C(x) = 1011 \rightarrow r = 3 \rightarrow 11010011000$. XOR steps:

bit 1: $1101 \oplus 1011 = 0110$ bit 2: $1100 \oplus 1011 = 0111$ bit 3: $1110 \oplus 1011 = 0101$ bit 4: $1011 \oplus 1011 = 0000$ bit 8: $1000 \oplus 1011 = 0011$ remainder = 011

R = 011 $\rightarrow$ transmitted **11010011 011**.

11. $11010001011 \div 1011 \rightarrow$ remainder **001** $\neq 0 \rightarrow$ **error detected** (bit 7 was flipped).

12. $C(x) = 1000 = x^3 \rightarrow R = 0$ exactly when the last 3 bits of M are 0 $\rightarrow M = xxx000$. Nonzero choices for the first 3 bits: $2^3 - 1 = 7$ values: **8, 16, 24, 32, 40, 48, 56**.

EX3 version: $C(x) = 100 \rightarrow$ last 2 bits 0 $\rightarrow 15$ values: 4, 8, ..., 60.

13. $L = 2 \cdot d \cdot R / c = 2 \times 1000 \times 10^8 / (2 \times 10^8) =$ **1000 bits**. Classic Ethernet: round trip $51.2\ \mu s \times 10\ Mb/s =$ **512 bits** (64 bytes). EX3: $2 \times 2500 \times 10^7 / 2 \times 10^8 = 250$ bits.

14. $n = 3 \rightarrow k \in \{0, \dots, 2^3 - 1 = 7\} \rightarrow$ wait $k \times 51.2\ \mu s$: 0, 51.2, 102.4, ..., **358.4 μs** (max).

15. (a) 00001000 00000000 00101011 11100100 10110001 00000010. (b) $0x012C = 256 + 32 + 12 =$ **300**. (c) Frames to its own address, the broadcast address (all 1s), and multicast addresses it was told to accept.

16. Hop 1 (H $\rightarrow$ R1): dst HW = R1; hop 2 (R1 $\rightarrow$ R2): dst HW = R2; hop 3 (R2 $\rightarrow$ D): dst HW = D. The IP addresses stay the same. H's HW address travels only **up to the closest router (R1)**.

17. (a) 1-persistent CSMA (Ethernet). (b) Non-persistent CSMA. (c) p-persistent CSMA. (d) ALOHA.
18. (a) **Hidden node**: A and C cannot hear each other, frames collide at B, neither knows. (b) **Exposed node**: C hears B and wrongly stays silent, though $C \rightarrow D$ would not disturb A. (c) Sender sends RTS, receiver replies CTS (both carry the length). Nodes that hear the **CTS** stay quiet (fixes hidden node); nodes that hear only the **RTS** may transmit (fixes exposed node).
19. ToDS = 1, FromDS = 1. Addr1 = **F** (ultimate destination), Addr2 = **AP-1** (immediate sender), Addr3 = **AP-2** (intermediate destination), Addr4 = **C** (original source).
20. $4 \text{ layers} \times 20\text{B} = 80\text{B}$: $1\text{B} \rightarrow 80/81 \approx \mathbf{0.988}$; $100\text{B} \rightarrow 80/180 \approx \mathbf{0.444}$. (If the application layer adds no header: $60/61$ and $60/160$.)

PART IV — JEOPARDY REVIEW (EX3 + MORE)

Questions 1–20 are EX3 with the professor's answers. 21–50 are new questions in the same style from the slides.

#	Question	Answer
1	If pkt dst is 3 hops away, whose HW@ is used as dst after hop 1?	Hop 2's HW@
2	Show an example of FSK of 10011	Figure: two frequencies, one for 1, one for 0
3	$B = 5000$ Hz, $S/N = 15$, max data rate?	20,000 b/s
4	Which 2B/3B codes do not give > two consecutive 0s?	010, 011, 101, 110, 111
5	$C(x) = 100$ for 6-bit $M(x)$: # of $M(x)$ with $R(x) = 0$?	15 values: 4, 8, ..., 60
6	Sublayers of the Data-Link layer?	LLC and MAC
7	Protocol that uses no carrier sense?	ALOHA
8	If channel is idle, wait random time — persistency type?	Non-persistent and p-persistent
9	$d = 2500$ m, $R = 10$ Mb/s, $c = 2 \times 10^8$ m/s: min L to detect collision?	250 bits
10	Hidden station problem in Wi-Fi?	Tx station close to the receiver but out of range of the other transmitter
11	Layers in OSI but not in TCP/IP?	Physical, Session, Presentation
12	Decimal value of 00:00:00:00:0A:00?	2560
13	R if $C = 1011$, $M = 11001010111001$?	$R = 0111$ (= 111), after appending 000
14	Sending a pkt to Instagram — how far does your phone's HW@ travel?	Up to your closest router
15	How to know 2 or more hosts are in the same LAN?	Same network ID (masked net address)
16	How does a packet network achieve resiliency to destruction?	Metadata + packet routing/forwarding
17	An example of an Internet killer application	E-mail, WWW
18	Early adopter institutions of the Internet (4)?	U. Utah, Stanford Research Institute, UCLA, UCSB
19	Why did Sputnik push the development of communications?	To make communications tolerant to partial destruction
20	Main function(s) of the Transport layer?	Reliable message delivery, flow control, error recovery
21	What does CSMA/CD stand for?	Carrier Sense Multiple Access with Collision Detection
22	Root protocol of Ethernet, and where was it developed?	ALOHA, University of Hawaii
23	Where/when was Ethernet developed? Which IEEE standard?	Xerox PARC, mid-1970s; IEEE 802.3
24	Max repeaters between two hosts; total reach?	4 repeaters; 2500 m
25	What does 10Base2 mean?	10 Mb/s, baseband, segment ≤ 200 m

26	10BaseT: cable, length, device?	Twisted pair, 100 m, hub (multiway repeater)
27	Encoding used by Ethernet?	Manchester
28	Ethernet frame fields and sizes?	Preamble 64, dst 48, src 48, type 16 bits, data 46–1500 B, CRC 32
29	Minimum Ethernet frame and why?	512 bits = 64 B; long enough to detect a collision (51.2 μ s round trip)
30	Size of the jam sequence; size of a runt frame?	32 bits; 96 bits (64 preamble + 32 jam)
31	Ethernet persistency type?	1-persistent
32	Exponential backoff after n collisions?	Wait $k \times 51.2 \mu\text{s}$, k random in $0 \dots 2^n - 1$
33	Ethernet address size, and who owns it?	48 bits; the adaptor (burned in ROM)
34	Broadcast address? Multicast address?	All 1s; first bit 1 but not all 1s
35	How many bits identify the manufacturer?	24-bit prefix (e.g. AMD 8:0:20)
36	p-persistent CSMA when idle?	Send with probability p, defer to next slot with $q = 1 - p$
37	Two spread spectrum techniques?	Frequency hopping, direct sequence (chipping code)
38	Restrictions on license-exempt bands?	Power limit + spread spectrum
39	Who allocates frequency bands in the USA?	FCC
40	802.11b / 802.11a / 802.11g speeds and bands?	11 Mb/s 2.4 GHz / 54 Mb/s 5 GHz OFDM / 54 Mb/s 2.4 GHz OFDM
41	Exposed node problem?	A node hears a transmission and wrongly holds back, although its own send would not interfere
42	What do RTS and CTS stand for?	Request To Send, Clear To Send
43	Node hears CTS — may it send? Hears only RTS?	No (close to receiver); yes (free to send)
44	How does 802.11 learn about a collision?	No CTS received $\rightarrow$ backoff (no collision detection)
45	Active vs passive scanning?	Active: node sends Probe; passive: AP sends Beacon
46	Frames to join an AP?	Probe, Probe Response, Association Request, Association Response
47	How many addresses in an 802.11 frame? Max data?	4 addresses; 2312 bytes
48	Bluetooth band, range, network name?	2.45 GHz, ~10 m, piconet (PAN)
49	Piconet size?	1 master + up to 7 slaves
50	ZigBee standard and goal?	IEEE 802.15.4; very low power, low bandwidth, cheap

PART V — EX3 (JEOPARDY) — ORIGINAL QUESTIONS & ANSWERS

The 20 questions from ECE422-EX3-J-Fall 2026 with the professor's answers, plus a one-line explanation.

1. If pkt dst 3 hops away, whose HW@ used as dst after hop 1?

A: Hop 2 HW@

Why: Each hop uses the next hop's hardware (MAC) address; IP addresses stay the same.

2. Show example of FSK of 10011

A: Show figure

Why: Constant amplitude; one frequency for 1, another for 0 (see Part III, Q2).

3. If $B = 5000$ Hz, $S/N = 15$, max data rate?

A: 20,000 b/s

Why: $C = 5000 \cdot \log_2(1+15) = 5000 \cdot \log_2 16 = 5000 \times 4$.

4. Which 2B/3B codes do not give > two consecutive 0s?

A: 010, 011, 101, 110, 111

Why: Drop 000, 001 (starts with 00) and 100 (ends with 00): at a boundary at most 1 + 1 zeros meet.

5. If $C(x) = 100$ for 6-bit $M(x)$, # of $M(x) \rightarrow R(x) = 0$?

A: 15 values: 4, 8, ..., 60

Why: $C(x) = x^2$ divides M exactly when the last 2 bits are 00 $\rightarrow$ xxxx00, first 4 bits nonzero: $2^4 - 1 = 15$.

6. Sublayers of the Data-Link layer?

A: LLC and MAC

Why: Logical Link Control + Media Access Control.

7. Protocol that uses no carrier sense?

A: ALOHA

Why: Pure ALOHA transmits at arbitrary times.

8. If channel is idle, wait random time. Persistency type?

A: Non-persistent and p-persistent

Why: 1-persistent never waits a random time before sending.

9. $d = 2500$ m, $R = 10$ Mb/s, $c = 2 \times 10^8$ m/s: min L to detect collision?

A: 250 bits

Why: $L = 2dR/c = 2 \times 2500 \times 10^7 / (2 \times 10^8) = 250$.

10. What is the hidden station problem in Wi-Fi?

A: Tx station close to the receiver but out of range of the other transmitter

Why: A and C both reach B but not each other $\rightarrow$ collision at B, neither detects it.

11. Layers in OSI but not in TCP/IP?

A: Physical, Session, Presentation

Why: TCP/IP (per slides): Application, Transport, Network, Link.

12. What decimal number is 00:00:00:00:0A:00?

A: 2560

Why: $0x0A00 = 10 \times 256 = 2560$.

13. R? if C = 1011, M = 11001010111001

A: R = 0111 (with +000)

Why: Append 3 zeros, XOR-divide by 1011 → remainder 111 (written 0111).

14. If transmitting a pkt to Instagram, how far does your phone's HW@ travel?

A: Up to your closest router

Why: MAC addresses are only used on one link; the router replaces them.

15. How to know 2 or more hosts are in the same LAN?

A: They have the same network ID (masked net @)

Why: Apply the mask to each IP address and compare.

16. How does a packet network achieve resiliency (to destruction)?

A: Using metadata and pkt routing/forwarding

Why: Each packet carries its addresses and can take another route.

17. An example of an Internet killer application

A: E-mail, WWW

Why: E-mail 1971, WWW 1991 (CERN).

18. Early adopter institutions of the Internet (4)?

A: U. Utah, Stanford R.I., UCLA, UCSB

Why: ARPANET's first 4 nodes, 1969.

19. Why did Sputnik push the development of communications?

A: To make communications tolerant to partial destruction

Why: 1957 → ARPA → packet switching, no single point of failure.

20. Main function(s) of the Transport layer?

A: Reliable message delivery, flow control, and error recovery

Why: End-to-end delivery between hosts.